

Components in wireless technology, 5XTC0

Module 5

Exercise: Design of amplifiers and low-noise amplifiers

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Where innovation starts

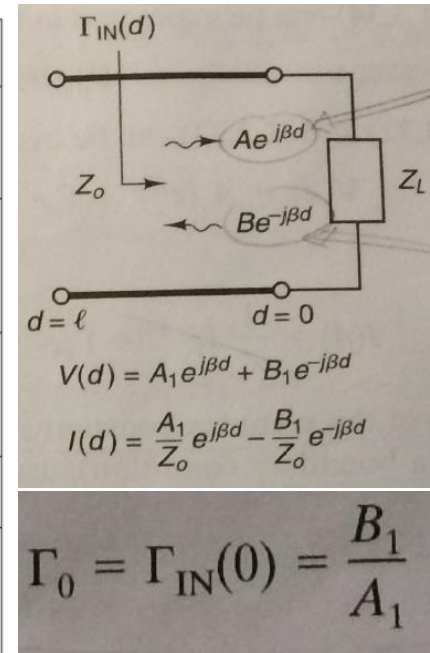
Outline

- Matching networks
- Amplifier design
- Low-noise amplifier design

Formula list available for exam

Power delivered to the load	$P_L = \frac{ V_2 ^2}{2Z_0} (1 - \Gamma_L ^2)$
Input power to the network	$P_{in} = \frac{ V_1 ^2}{2Z_0} (1 - \Gamma_{in} ^2)$
Input and output reflection coefficients of a transistor with a source and load: general case	$\Gamma_{in} = \frac{V_1^-}{V_1^+} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$ $\Gamma_{out} = \frac{V_2^-}{V_2^+} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S}$
Input and output reflection coefficients of a transistor with a source and load: unilateral case	$\Gamma_{in} = \frac{V_1^-}{V_1^+} = S_{11}$ $\Gamma_{out} = \frac{V_2^-}{V_2^+} = S_{22}$
Gain of the input matching network	$G_S = \frac{1 - \Gamma_S ^2}{ 1 - \Gamma_{in}\Gamma_S ^2}$
Gain of the output matching network	$G_L = \frac{1 - \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2}$
Gain of the transistor (unilateral case)	$G_0 = S_{21} ^2$
Transducer gain of the basic amplifier circuit (input matching, unilateral transistor, output matching)	$G_T = G_S G_0 G_L$ $G_{T,db} = G_{S,db} + G_{0,db} + G_{L,db}$
Maximum gain of the input and output matching networks	$G_{S,max} = \frac{1}{1 - S_{11} ^2}$ $G_{L,max} = \frac{1}{1 - S_{22} ^2}$
Maximum transducer power gain, unilateral case	$G_{TU,max} = \frac{1}{1 - S_{11} ^2} S_{21} ^2 \frac{1}{1 - S_{22} ^2}$
Normalized gain factors g_s and g_L	$g_s = \frac{G_S}{G_{S,max}} = \frac{1 - \Gamma_S ^2}{ 1 - S_{11}\Gamma_S ^2} (1 - S_{11} ^2)$ $g_L = \frac{G_L}{G_{L,max}} = \frac{1 - \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2} (1 - S_{22} ^2)$
Center and radius of the constant gain circle for the input matching network	$C_S = \frac{g_S S_{11}^*}{1 - (1 - g_S) S_{11} ^2}$ $R_S = \frac{\sqrt{1 - g_S} (1 - S_{11} ^2)}{1 - (1 - g_S) S_{11} ^2}$
Center and radius of the constant gain circle for the output matching network	$C_L = \frac{g_L S_{22}^*}{1 - (1 - g_L) S_{22} ^2}$ $R_L = \frac{\sqrt{1 - g_L} (1 - S_{22} ^2)}{1 - (1 - g_L) S_{22} ^2}$
Condition for "unconditionally stable" device, general case	for all $ \Gamma_L < 1$ and $ \Gamma_S < 1$ $\Rightarrow \begin{cases} \Gamma_{in} = \left S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} \right < 1 \\ \Gamma_{out} = \left S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S} \right < 1 \end{cases}$

Conditions for "unconditionally stable" device, unilateral case	$ \Gamma_{in} = S_{11} < 1$ $ \Gamma_{out} = S_{22} < 1$
Center and radius of the stability circles, load side	$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{ S_{22} ^2 - \Delta ^2}$ (center), $R_L = \frac{ S_{12}S_{21} }{ S_{22} ^2 - \Delta ^2}$ (radius), $\Delta = S_{11}S_{22} - S_{12}S_{21}$
Center and radius of the stability circles, source side	$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{ S_{11} ^2 - \Delta ^2}$ (center), $R_S = \frac{ S_{12}S_{21} }{ S_{11} ^2 - \Delta ^2}$ (radius), $\Delta = S_{11}S_{22} - S_{12}S_{21}$
Test for unconditional stability, general case	$ \Delta = S_{11}S_{22} - S_{12}S_{21} < 1$ and $K = \frac{1 - S_{11} ^2 - S_{22} ^2 + \Delta ^2}{2 S_{12}S_{21} } > 1$
Test for unconditional stability, unilateral case	$ S_{11} < 1$ $ S_{22} < 1$
Two-stage amplifier: Output noise and noise figure	$P_{N,total} = G_{A2}(G_{A1}P_{N,in} + P_{n1}) + P_{n2}$ $\Rightarrow F_{total} = \frac{P_{N,total}}{P_{N,in}G_{A1}G_{A2}} = 1 + \frac{P_{n1}}{P_{N,in}G_{A1}} + \frac{P_{n2}}{P_{N,in}G_{A1}G_{A2}}$ $F_{total} = F_1 + \frac{F_2 - 1}{G_{A1}}$ with $F_j = 1 + \frac{P_{nj}}{P_{N,in}G_{Aj}}$, $j=1,2$
Noise figure of a 2-port amplifier	$F = F_{min} + \frac{P_N}{g_S} \left \frac{y_S - y_{opt}}{y_S - y_{opt}} \right ^2$ $F = F_{min} + 4r_N \frac{ \Gamma_S - \Gamma_{opt} ^2}{(1 - \Gamma_S ^2)(1 + \Gamma_{opt} ^2)}$
Constant noise circles	$C_F = \frac{\Gamma_{opt}}{1 + N}$ $R_F = \frac{1}{1 + N} \sqrt{N^2 + N(1 - \Gamma_{opt} ^2)}$ $\Delta F_n' = N = (F - F_{min}) \frac{1 + \Gamma_{opt} ^2}{4r_N} = \frac{ \Gamma_S - \Gamma_{opt} ^2}{1 - \Gamma_S ^2}$



Transmission lines: Problem 1

- Book of Gonzalez: Example 1.13
- Assignment, problem 2

- Calculate the reflection coefficient Γ_{IN} and the VSWR in the $\lambda/4$ line in Fig. 2.
- Calculate the voltage $V(\lambda/4)$, the power available from the source P_{AVS} , and the power delivered to the $100\ \Omega$ load.

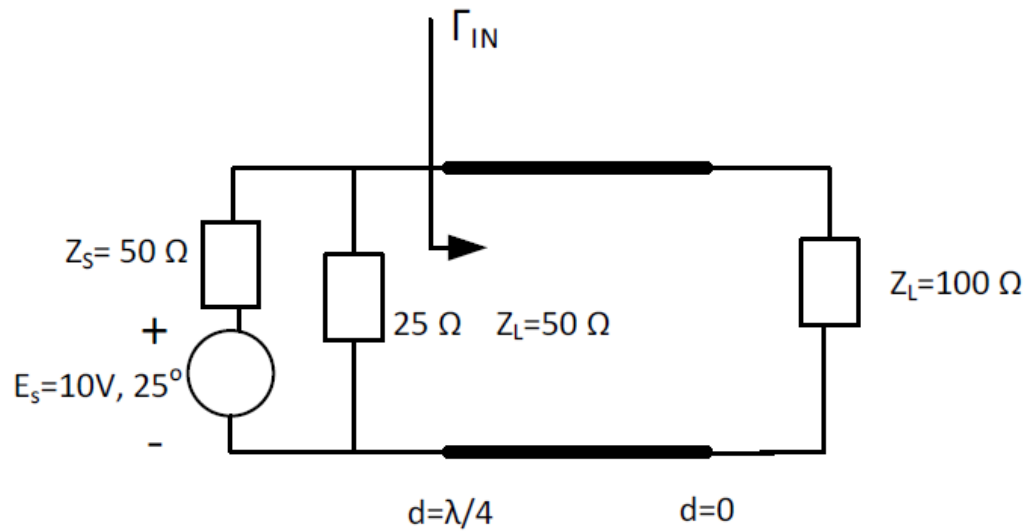
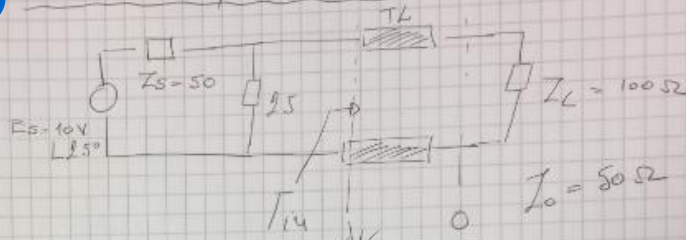


Figure 2 Problem 2.

Transmission lines: Solution for Problem 1a

a) Gonzalez, Problem 1.13



$$V = A_1 e^{j\beta d} + B_1 e^{-j\beta d}$$

$$V = A_1 e^{j\beta d} \left(1 + \frac{B_1}{A_1} e^{-j2\beta d} \right)$$

$$\Gamma_0 = \frac{B_1}{A_1}$$

$$\Gamma(d) = \frac{B_1}{A_1} e^{-j2\beta d} = \Gamma_0 e^{-j2\beta d}$$

$$V(d) = A_1 e^{j\beta d} (1 + \Gamma_0 e^{-j2\beta d})$$

$$VSWR = \frac{1 + |\Gamma_0|}{1 - |\Gamma_0|}$$

$$a) \Gamma_{in} = \Gamma(\lambda/4) = \Gamma_0 e^{-j\pi}$$

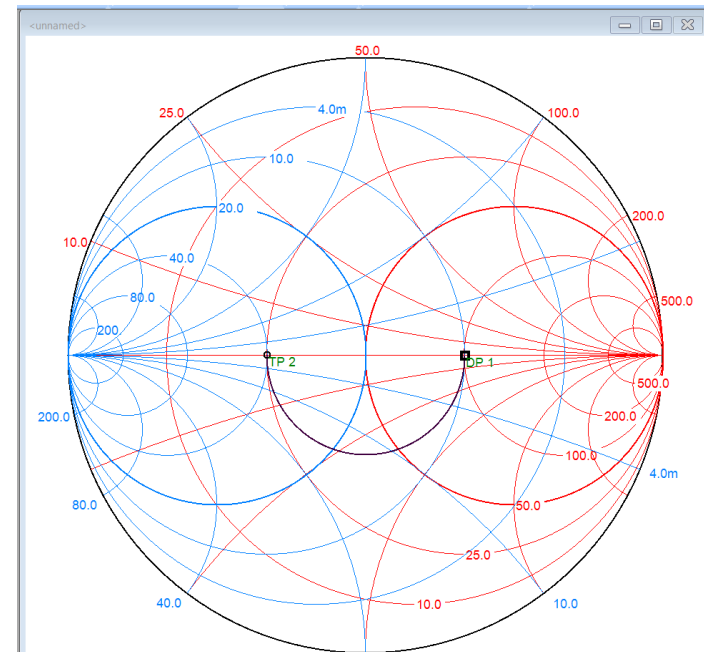
$$\Gamma_0 = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{100 - 50}{100 + 50} = \frac{50}{150} = 0.33$$

$$\Gamma_{in} = \Gamma(\lambda/4) = 0.33 e^{j2\pi \frac{\lambda}{4} \frac{\lambda}{4}}$$

$$\Gamma_{in} = 0.33 e^{-j\pi}$$

$$\Gamma_{in} = -0.33$$

$$VSWR = \frac{1 + 0.33}{1 - 0.33} = 1.98$$



Transmission lines: Solution for Problem 1 b (1/2)

$$b) \quad Z_{in} = Z_0 \frac{Z_L + j Z_0 \tan \beta d}{Z_0 + j Z_L \tan \beta d}$$

$$d = \frac{\lambda}{4}$$

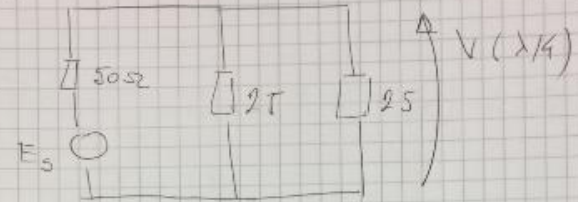
$$\beta d = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \pi/2$$

$$\tan \pi/2 = \infty$$

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

$$Z_{in} = \frac{50^2}{100} = 25 \Omega$$

Equivalent circuit:



$$V(\frac{\lambda}{4}) = E_s \cdot \frac{12.5}{50 + 12.5}$$

$$V(\frac{\lambda}{4}) = 10 e^{j2.5} \cdot 0.2$$

$$V(\frac{\lambda}{4}) = 2 e^{j2.5}$$

$$V(\frac{\lambda}{4}) = A_1 e^{j\beta d} (1 + \Gamma_0 e^{-j2\beta d})$$

$$\frac{\lambda}{4} = d \Rightarrow \beta d = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$$

$$2\beta d = \pi$$

$$2 \cdot e^{j2.5} = A_1 e^{j\frac{\pi}{2}} (1 + 0.33 e^{-j\pi})$$

$$2 e^{-j6.5} = A_1 (1 - 0.33)$$

$$A_1 = 3 e^{-j6.5}$$

$$e^{-j\pi} = \cos(\frac{-\pi}{2} + j\frac{\pi}{2})$$

$$e^{-j\pi} = -1$$

Transmission lines: Solution for Problem 1 b (2/2)



$$P_{avs} @ Z_L = Z_S^* = \sqrt{P_o} \left| V_L \cdot I_L^* \right|$$

$$Z_S = Z_S + jX_S$$

$$Z_L = Z_S^* = Z_S - jX_S$$

$$V_L = \frac{E_S}{2}$$

$$I_L = \frac{E_S}{2Z_S}$$

$$P_{avs} = \frac{|E_S|^2}{8Z_S} = \frac{100}{8 \cdot 50} = 0,25 \text{ W}$$

$$V(o) = A_1 (1 + \Gamma_o)$$

$$V(o) = 3 e^{-j65} (1 + 0,33) = 3,99$$

$$V(o) = 3,99 e^{-j65}$$

$$I(o) = \frac{A_1 (1 - \Gamma_o)}{Z_o} = \frac{3 e^{-j65} \cdot 0,67}{50}$$

$$I(o) = 0,04 e^{-j65}$$

$$P_o = \frac{\text{Re}}{2} \left| V(o) \cdot I(o)^* \right|$$

max. rms

$$P_o = \frac{\text{Re}}{2} \left| 3,99 e^{-j65} \cdot 0,04 e^{j65} \right|$$

$$P_o = 0,08 \text{ W}$$

Matching: Problem 2

- Assignment, problem 3

Design an LC matching network that transforms impedance $Z=100+j50$ to 50Ω .
Operating frequency is 1GHz.

- How many solutions are existing?
Draw the solutions in the Smith Chart
- Calculate impedance of L and C for all solutions
- Calculate values of L and C for all solutions
- Take one of the solutions and check solution based on Smith Chart with analytical calculation

Series L with impedance $Z_l = j1.2 \cdot 50 = j60 \text{ ohm}$
 $j \cdot 2 \cdot \pi \cdot f \cdot L = j60$
 $L = 1 / (2 \cdot \pi \cdot f \cdot 60) = 9.5 \text{ nH}$

Series C with impedance $Z_c = -j1.2 \cdot 50 = -j60 \text{ ohm}$
 $-j \cdot 1 \cdot (2 \cdot \pi \cdot f \cdot C) = -j60$
 $C = 1 / (2 \cdot \pi \cdot f \cdot 60) = 2.6 \text{ pF}$

Shunt L with admittance $Y_l = (-j0.5 + j0.2) \cdot 0.02 = -j0.006$
 $Z_l = 1 / Y_l = 166.6 \text{ ohm}$
 $-j \cdot 1 / (2 \cdot \pi \cdot f \cdot L) = -j0.006$
 $L = 1 / (0.006 \cdot 2 \cdot 3.14 \cdot 1e9) = 26.5 \text{ nH}$

Shunt C with admittance $Y_c = (j0.5 + j0.2) \cdot 0.02 = j0.006$
 $Z_c = 1 / Y_c = j71.4 \text{ ohm}$
 $-j \cdot 1 / (2 \cdot \pi \cdot f \cdot C) = j71.4$
 $C = 1 / (71.4 \cdot 2 \cdot 3.14 \cdot 1e9) = 2.2 \text{ pF}$

RADIALLY SCALED PARAMETERS

TOWARD LOAD →

← TOWARD GENERATOR

SWR

dBS

REFL. COEFF. P

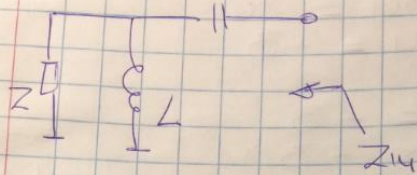
REFL. COEFF. V

TRANSM. COEFF. P

TRANSM. COEFF. V

CENTER

Matching: Problem 2 – Solution (2/2)



$$Z_{in} = \frac{1}{j\omega C} + \frac{1}{\frac{1}{j\omega L} + \frac{1}{Z}}$$

$$Z_{in} = -j \frac{1}{\omega C} + \frac{1}{-j \frac{1}{\omega L} + \frac{1}{Z}}$$

$$\frac{1}{\omega C} = A, \quad \frac{1}{\omega L} = B \quad \dots (1)$$

$$Z_{in} = -j \cdot A + \frac{1}{-jB + Z + jI_m}$$

$$\frac{1}{Z} = Z + jI_m \rightarrow \text{known} \quad \left\{ \begin{array}{l} Z = 100 + j50 \\ B = 8 \cdot 10^{-3}, I_m = 4 \cdot 10^{-3} \end{array} \right.$$

$$Z_{in} = -j \cdot A + \frac{R - j(I_m - B)}{R^2 + (I_m - B)^2}$$

$$Z_{in} = 50 \rightarrow \text{known}$$

$$50 = \frac{R}{R^2 + (I_m - B)^2} \quad \dots (3)$$

$$A + \frac{I_m - B}{R^2 + (I_m - B)^2} = 0 \quad \dots (4)$$

$$(I_m - B)^2 = \frac{R}{50} - R^2$$

$$B = \pm \sqrt{\frac{R}{50} - R^2} + I_m$$

$$B = 5,8 \cdot 10^{-3} \Rightarrow L = 27,45 \mu H$$

$$\text{from (1)} \Rightarrow L = \frac{1}{2 \cdot \pi \cdot f \cdot B}$$

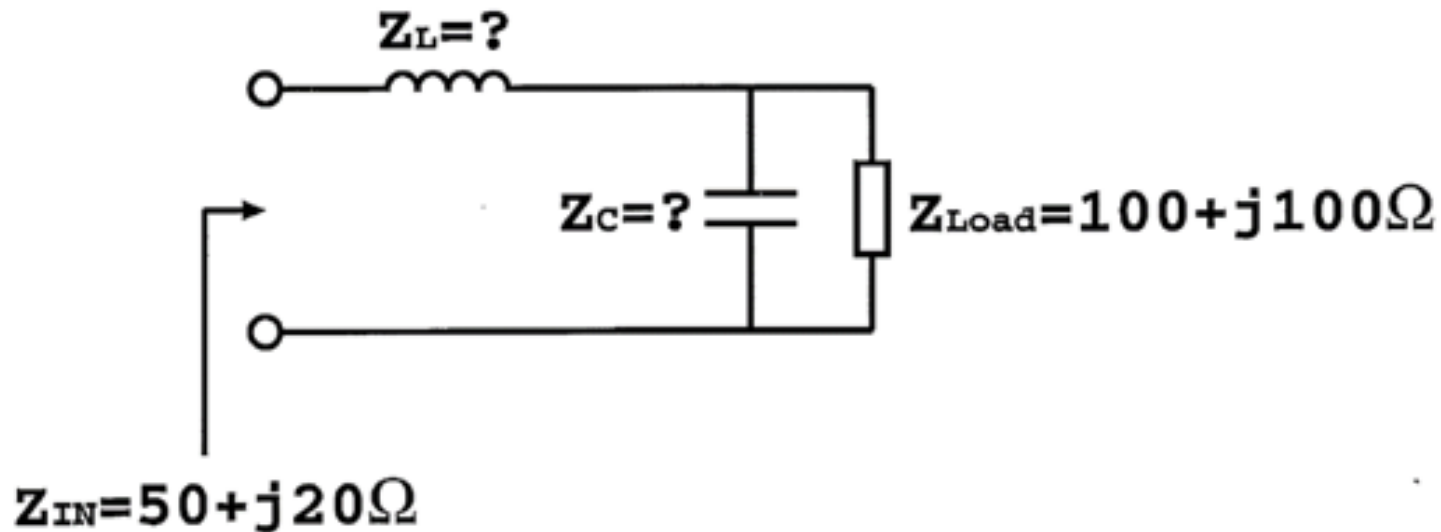
$$A = \frac{B - I_m}{R^2 + (I_m - B)^2}$$

$$A = 61,23 \Rightarrow C = \frac{1}{2 \cdot \pi \cdot f \cdot A} =$$

$$C = 2.6 pF$$

Matching: Problem 3

Using the Smith chart, design a matching network with lumped elements (L and C) as shown below which transforms the load impedance Z_{Load} into the input impedance Z_{IN} .



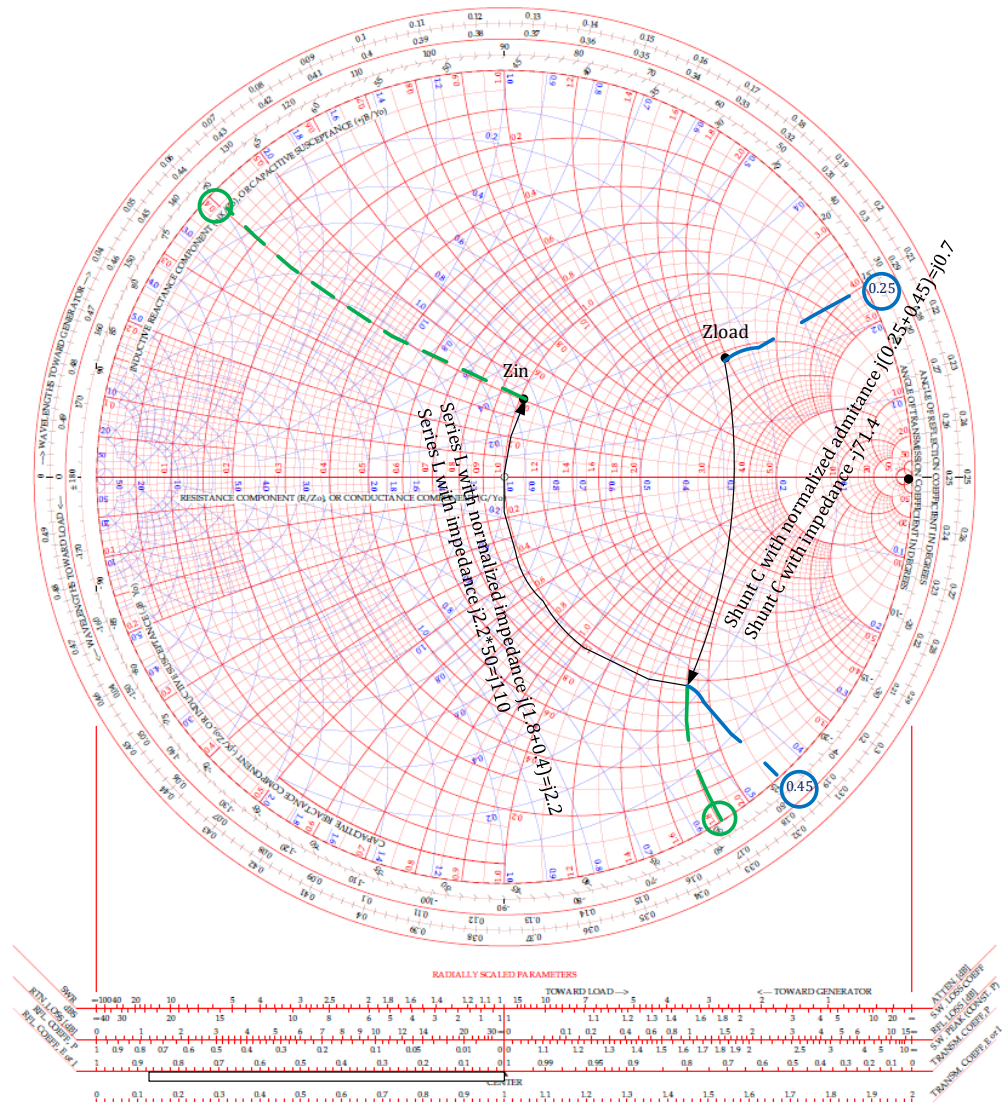
Problem 3: solution

Shunt C with normalized admittance $j(0.25+0.45)=j0.7$

Shunt C with impedance $(1/j0.7)*50=-j71.4$

Series L with normalized impedance $j(1.8+0.4)=j2.2$

Series L with impedance $j2.2*50=j110$



Amplifier design:

Similar as example 12.1 book of Pozar

A microwave transistor has the following S parameters at 10 GHz, with a $50\ \Omega$ reference impedance:

$$S_{11} = 0.45 \angle 150^\circ$$

$$S_{12} = 0.01 \angle -10^\circ$$

$$S_{21} = 2.05 \angle 10^\circ$$

$$S_{22} = 0.40 \angle -150^\circ$$

The source impedance is $Z_S = 20\ \Omega$ and the load impedance is $Z_L = 30\ \Omega$. Compute the power gain, the available gain, and the transducer power gain.

Amplifier design

Similar as example 12.1 book of Pozar - Solution

Example 11.1, book of Pozar, page 539

$$S_{11} = 0.45 \angle 150^\circ = 0.45 e^{j150^\circ}$$

$$S_{12} = 0.01 e^{j10^\circ}$$

$$S_{21} = 2.05 e^{j10^\circ}$$

$$S_{22} = 0.4 e^{-j150^\circ}$$

$$Z_S = 20 \Omega$$

$$Z_L = 30 \Omega$$

$$Z_0 = 50 \Omega$$

$$\text{Power gain} = G = \frac{|S_{21}|^2 (1 - |Z|^2)}{(1 - |T_{11}|^2) (1 - |S_{22}|^2)}$$

$$\text{Available power gain} = G_A = \frac{|S_{21}|^2 (1 - |Z_S|^2)}{|1 - S_{11} Z_S|^2 (1 - |T_{out}|^2)}$$

$$\text{Transducer power gain} = G_T = \frac{|S_{21}|^2 (1 - |Z_S|^2) (1 - |Z_L|^2)}{|1 - T_S T_{11}|^2 |1 - S_{22} Z_L|^2}$$

[S] parameters are known

$$T_L = ?, T_S = ?, T_{out} = ?, T_{11} = ?$$

$$T_L = \frac{Z_L - Z_0}{Z_L + Z_0} = -0.150$$

$$T_S = \frac{Z_S - Z_0}{Z_S + Z_0} = -0.429$$

$$T_{out} = S_{22} + \frac{S_{12} S_{21} T_{L5}}{1 - S_{11} T_S}$$

$$T_{11} = S_{11} + \frac{S_{12} S_{21} T_L}{1 - S_{22} T_L}$$

$$T_{out} = 0.408 e^{-j151^\circ}$$

$$T_{11} = 0.455 e^{j150^\circ}$$

$$G = 5.99$$

$$G_A = 5.85$$

$$G_T = 5.49$$

Amplifier design:

Example 3.4.1 book of Gonzalez

The S parameters of a BJT measured at $V_{CE} = 10$ V, $I_C = 30$ mA, and $f = 1$ GHz, in a $50\text{-}\Omega$ system are

$$S_{11} = 0.73 \angle 175^\circ$$

$$S_{12} = 0$$

$$S_{21} = 4.45 \angle 65^\circ$$

$$S_{22} = 0.21 \angle -80^\circ$$

- (a) Calculate the optimum terminations.
- (b) Calculate $G_{s,\max}$, $G_{L,\max}$, and $G_{TU,\max}$ in decibels.
- (c) Draw several G_s constant-gain circles.
- (d) Design the input matching network for $G_s = 2$ dB.

Example 3.4.1 book of Gonzalez – Solution (1/3)

Example 3.4.1, book of Gonzalez, page 132

$f = 1 \text{ GHz}$

$$S_{11} = 0.73 e^{j175}$$

$S_{12} = 0 \rightarrow$ unilateral case

$$S_{21} = 4.45 e^{j65}$$

$$S_{22} = 0.21 e^{j-80}$$

$$G_T = \underbrace{\frac{1 - |\Gamma_S|^2}{|1 - \Gamma_S \Gamma_{in}|^2}}_{G_S} \underbrace{|S_{21}|^2}_{G_0} \underbrace{\frac{1 - |\Gamma_L|^2}{|1 - S_{22} \Gamma_L|^2}}_{G_L}$$

1) $G_{T \max}$ / Conjugate match at input, output
 $\Gamma_{in} = \Gamma_S^*$, $\Gamma_{out} = \Gamma_L^*$

$$G_{S \max} = \frac{1 - |\Gamma_S|^2}{(1 - |\Gamma_S|^2)^2} = \frac{1}{1 - |\Gamma_S|^2}$$

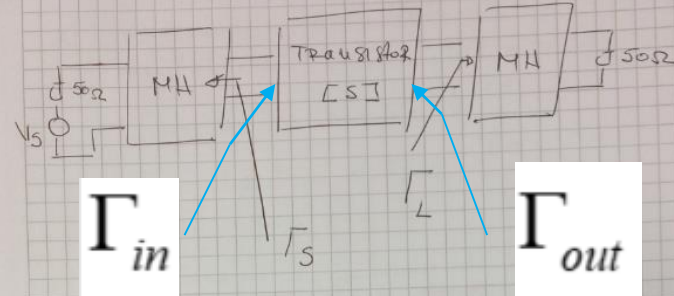
$$G_{T \max} = \frac{1}{1 - |\Gamma_S|^2} |S_{21}|^2 \frac{1 - |\Gamma_L|^2}{|1 - S_{22} \Gamma_L|^2}$$

$$\Gamma_S^* = S_{11} + \frac{S_{12} S_{21} \Gamma_L}{1 - S_{22} \Gamma_L}$$

$$\Gamma_L^* = S_{22} + \frac{S_{12} S_{21} \Gamma_S}{1 - S_{11} \Gamma_S}$$

$$\Gamma_{in} = \frac{V_1^-}{V_1^+} = S_{11} + \frac{S_{12} S_{21} \Gamma_L}{1 - S_{22} \Gamma_L}$$

$$\Gamma_{out} = \frac{V_2^-}{V_2^+} = S_{22} + \frac{S_{12} S_{21} \Gamma_S}{1 - S_{11} \Gamma_S}$$



2) $G_{T \max}$ / unilateral case $S_{12} = 0$
 Conjugate match at input and output
 $\Gamma_{in} = \Gamma_S^*$, $\Gamma_{out} = \Gamma_L^*$

$$\Gamma_S^* = S_{11} \Rightarrow \Gamma_S = S_{11}^*$$

$$\Gamma_L^* = S_{22} \Rightarrow \Gamma_L = S_{22}^*$$

$$G_{T \max} = \frac{1}{1 - |S_{11}|^2} |S_{21}|^2 \frac{1}{1 - |S_{22}|^2}$$

$$G_{S \max} = \frac{1}{1 - |S_{11}|^2}$$

$$G_{L \max} = \frac{1}{1 - |S_{22}|^2}$$

Example 3.4.1 book of Gonzalez – Solution (2/3)

a) optimum terminations $\rightarrow$
unilateral case $S_{12} = 0 \rightarrow$

$$\Gamma_S = S_{11}^* = 0.73 e^{-j175}$$

$$\Gamma_L = S_{22}^* = 0.21 e^{+j80}$$

b) $G_{S_{11} \max} = \frac{1}{1 - |S_{11}|^2} = 2.141$

$$G_{S_{11} \max} [dB] = 10 \log G_{S_{11} \max} = 3.3 dB$$

$$G_{S_{22} \max} = \frac{1}{1 - |S_{22}|^2} = 1.046$$

$$G_{S_{22} \max} [dB] = 0.195$$

$$G_0 = |S_{21}|^2 = 19.8$$

$$G_0 [dB] = 12.97 dB$$

$$G_{TU, \max} = 16.47 dB = G_{S_{11} \max} + G_0 + G_{S_{22} \max} \rightarrow \text{all in dB}$$

c) $G_S = 2 dB \rightarrow$ in dB

$$G_S = 1.59$$

$$g_S = \frac{G_S}{G_{S_{11} \max}} = \frac{1.59}{2.141} = 0.74$$

$$C_S = \frac{g_S \cdot S_{11}^*}{1 - (1 - g_S) |S_{11}|^2}$$

$$R_S = \frac{\sqrt{1 - g_S} (1 - |S_{11}|^2)}{1 - (1 - g_S) |S_{11}|^2}$$

$$C_S = 0.625 e^{j-175}$$

$$R_S = 0.274$$

[illegible]

Amplifier design: problem 4

A microwave amplifier design is to be designed for $G_{TU,max}$ using a transistor with

$$S_{11}=0.5, \angle 140^\circ$$

$$S_{12} = 0$$

$$S_{21}=5, \angle 45^\circ$$

$$S_{22}=0.6, \angle -95^\circ$$

The S-parameters were measured in a $50\ \Omega$ system at $f=900\text{ MHz}$, $V_{CE} = 15\text{ V}$, and $I_C = 15\text{ mA}$.

- Determine whether this amplifier is unconditionally stable.
- Determine $G_{TU,max}$. Express your answer in dB.
- Determine the optimum source impedance and the optimum load impedance for maximum gain. Illustrate your answer in the Smith chart.
- Design a 2-element matching network at the input and a 2-element matching network at the output of the amplifier to reach conjugate matching to a $50\ \Omega$ source and load impedance. Illustrate your answer on the Smith chart.
- Calculate the constant gain circle for $G_L=1\text{dB}$ and indicate it in the Smith chart.

Amplifier design: Problem 4 - solution (1/5)

Problem 1

$$S_{11} = 0.5 e^{j40}$$

$$S_{12} = 0$$

$$S_{21} = 5 e^{j45}$$

$$S_{22} = 0.6 e^{-j95}$$

$$a) \Delta = S_{11} S_{22} - S_{12} S_{21} < 1$$

$$K_0 = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + \Delta^2}{2 |S_{12} S_{21}|} > 1 \quad \left. \begin{array}{l} \text{general} \\ \text{case} \end{array} \right\}$$

$$|S_{11}| < 1$$

$$|S_{22}| < 1 \quad \left. \begin{array}{l} \text{unilateral case} \end{array} \right\}$$

Conditions for unconditional stability

$S_{12} = 0$, unilateral case

$|S_{11}| = 0.5 < 1$, $|S_{22}| = 0.6 < 1 \rightarrow$ amplifier
unconditionally
stable

$$b) G_{TUMAX} = \frac{1}{1 - |S_{11}|^2}$$

G_{TUMAX} / unilateral case $S_{12} = 0$

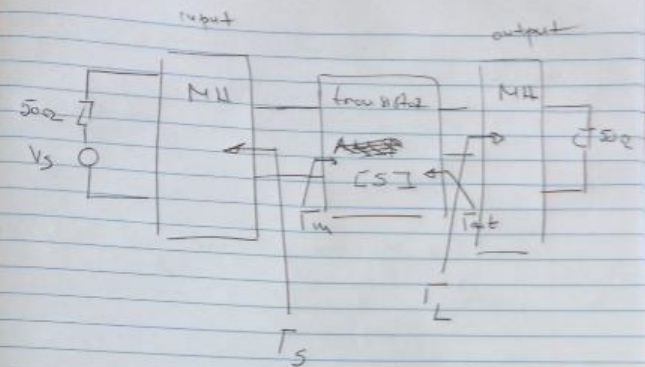
$$G_{TUMAX} = \frac{1}{1 - |S_{11}|^2} \frac{1}{1 - |S_{22}|^2}$$

$$\Gamma_S = S_{11}^*$$

$$\Gamma_L = S_{22}^*$$

(1)

1.30



$$G_{TUMAX} = \frac{1}{1 - |S_{11}|^2} \frac{1}{1 - |S_{22}|^2}$$

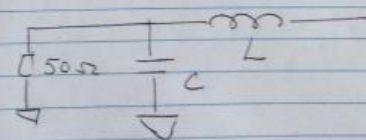
$$G_{TUMAX} = 1.33 \cdot 25 \cdot 1.56 = 52$$

$$G_{TUMAX} [dB] = 10 \log G_{TUMAX} = 17.16 \text{ dB}$$

$$c) \Gamma_S = S_{11}^* = 0.5 e^{-j40}$$

$$\Gamma_L = S_{22}^* = 0.6 e^{j95}$$

d) M1 @ input



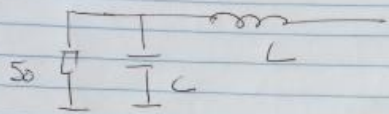
$$C = 6.7 \text{ pF}$$

$$L = 1.44 \text{ nH}$$

(2)

Amplifier design: Problem 4 - solution (2/5)

Matching at output



$$C = 4 \text{ pF}$$

$$L = 11.7 \text{ nH}$$

e) $G_L = 1 \text{ dB} \Rightarrow G_L = 1.26$

$$G_{L\max} = \frac{1}{1 - |S_{22}|^2} = 1.56$$

$$g_L = \frac{G_L}{G_{L\max}} = 0.8$$

$$C_L = \frac{g_L \cdot S_{22}^*}{1 - (1 - g_L) |S_{22}|^2}$$

$$P_L = \frac{\sqrt{1 - g_L} (1 - |S_{22}|^2)}{1 - (1 - g_L) |S_{22}|^2}$$

$$C_L = \frac{0.8 \cdot 0.6 e^{j95}}{1 - (1 - 0.8) |0.6|^2}$$

$$C_L = \frac{0.48 e^{j95}}{0.928}$$

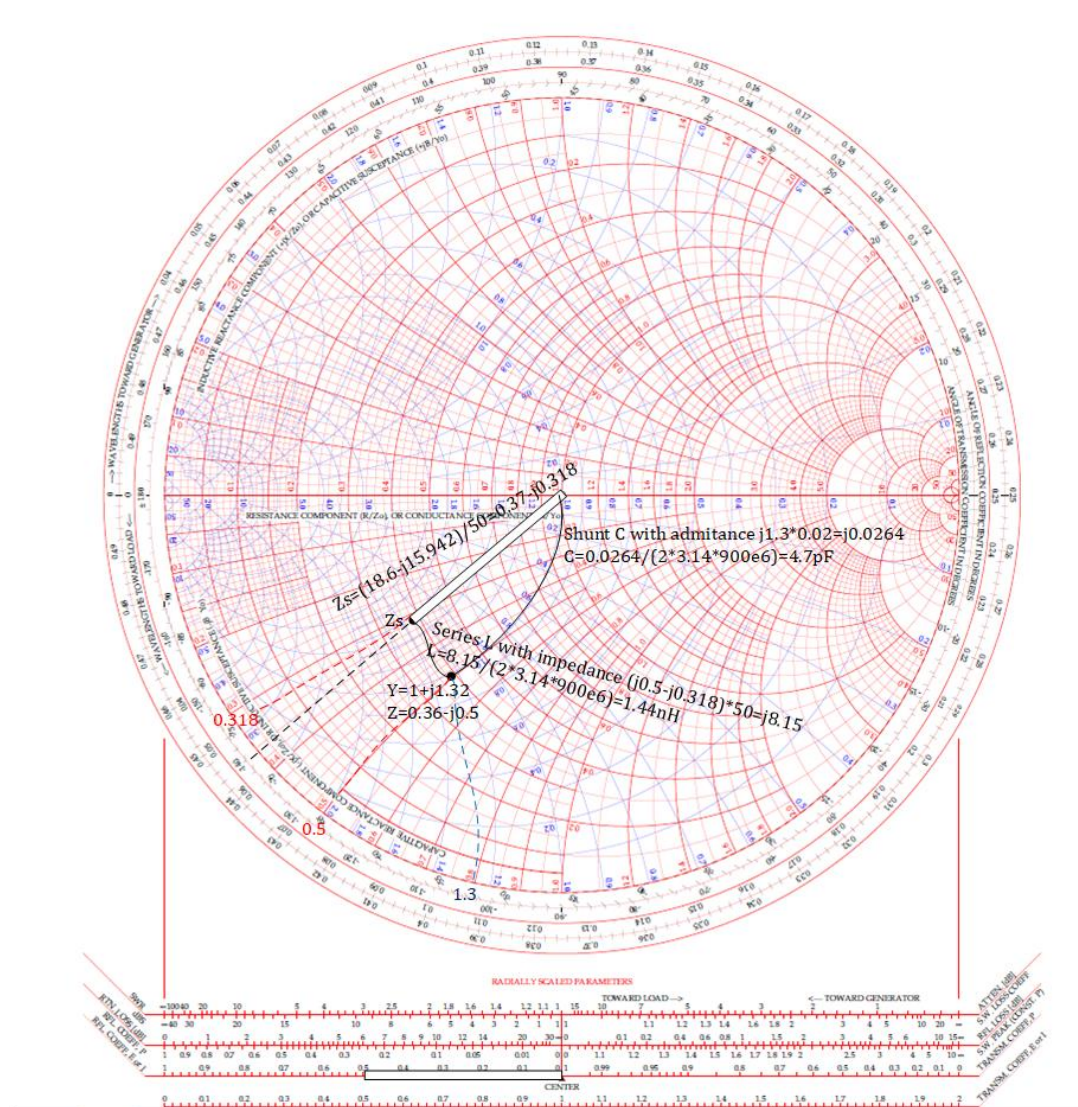
$$C_L = 0.52 e^{j95}$$

$$P_L = \frac{\sqrt{1 - 0.8} (1 - 0.6^2)}{1 - (1 - 0.8) (0.6)^2}$$

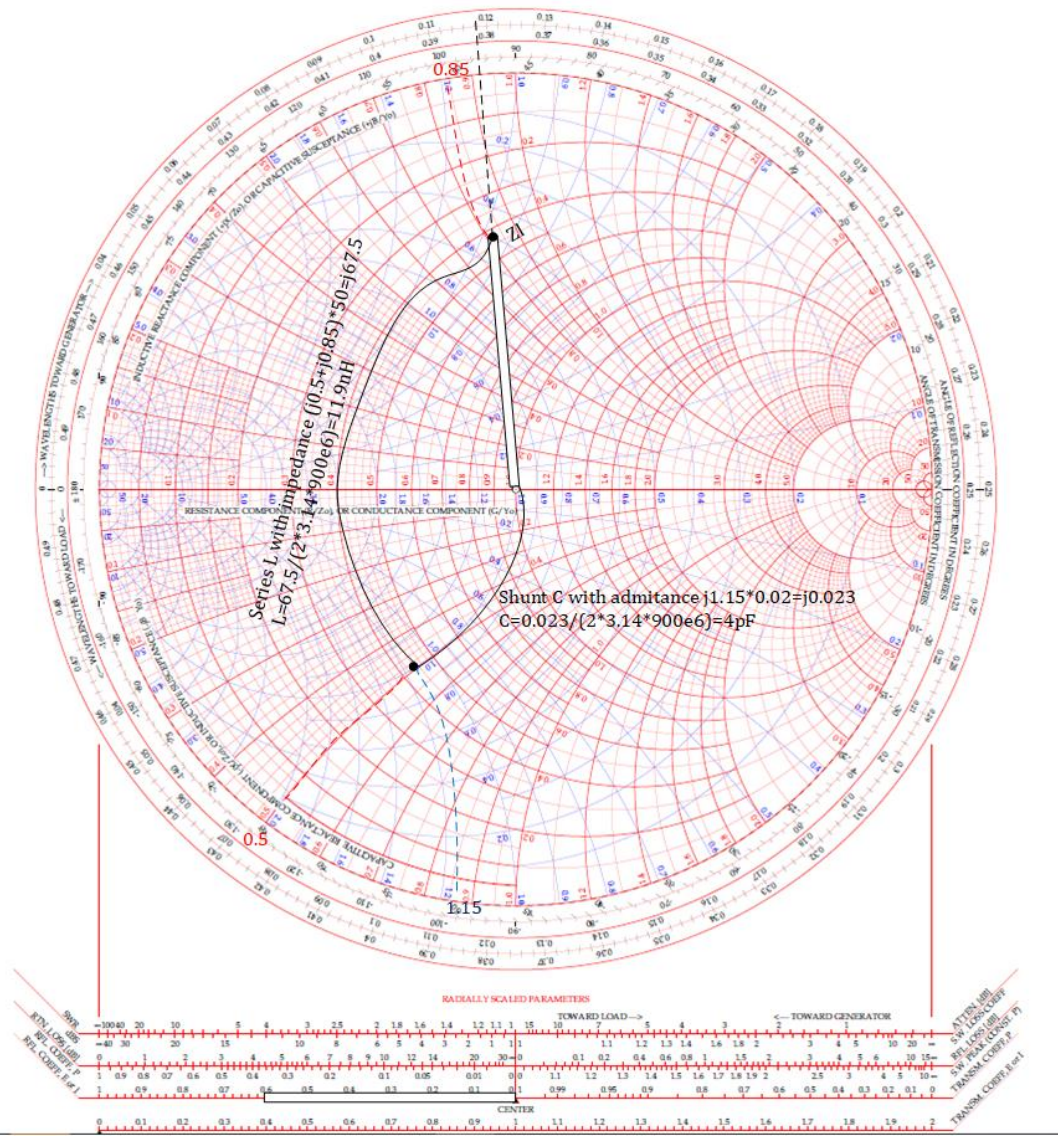
$$P_L = \frac{0.286}{0.928}$$

$$P_L = 0.31$$

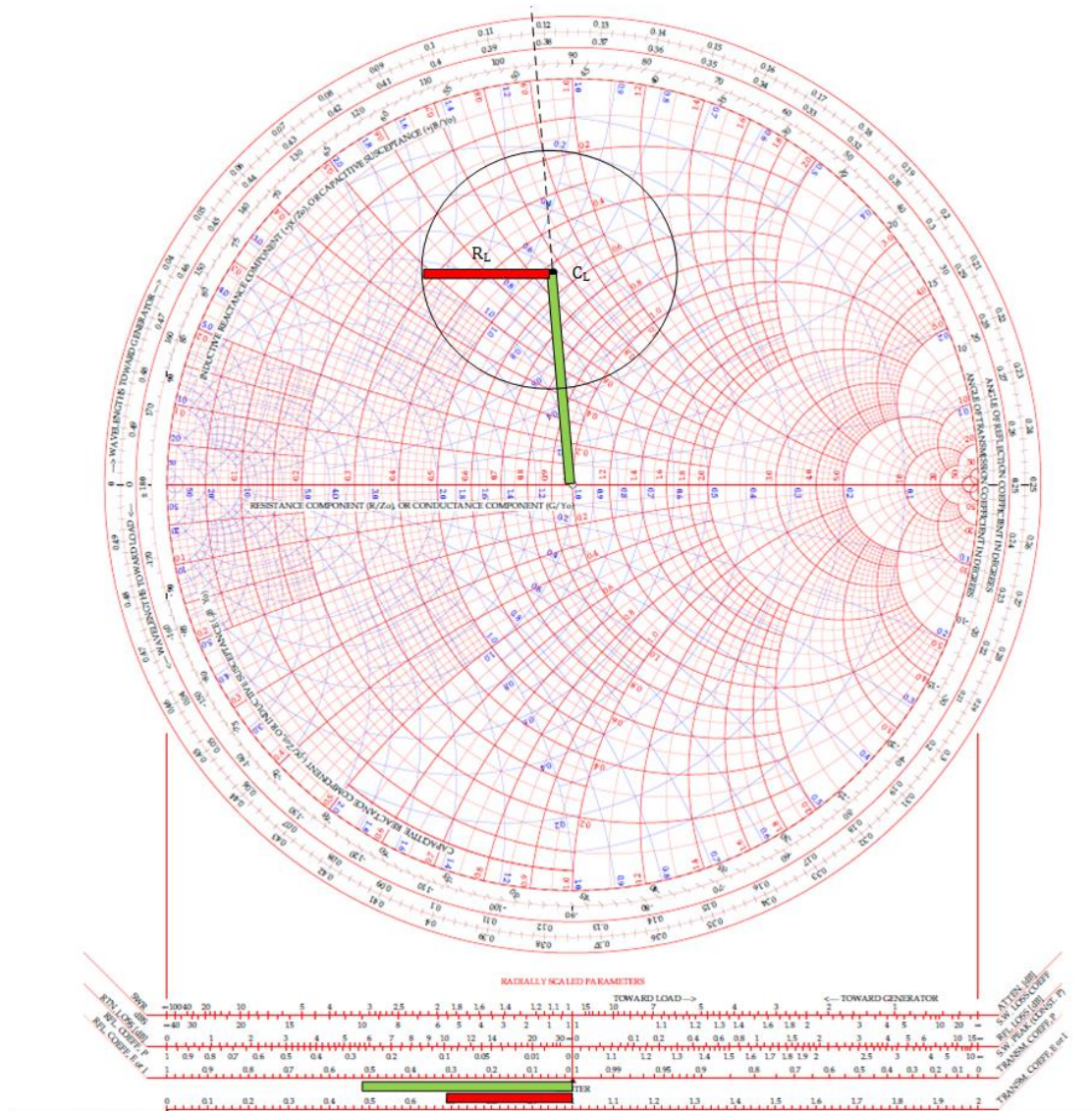
Amplifier design: Problem 4 - solution, input matching (3/5)



Amplifier design: problem 4 - solution, output matching (4/5)



Amplifier design: problem 4 - solution, output matching (5/5)



Problem 5 – will be in assignment

A microwave amplifier is to be designed for $G_{TU,max}$ using a bipolar transistor with:

$$S_{11}=0.4, \angle 130^\circ$$

$$S_{12} = 0$$

$$S_{21}=6, \angle 40^\circ$$

$$S_{22}=0.7, \angle -75^\circ$$

The S-parameters were measured in a $50\ \Omega$ system at $f=1\text{ GHz}$, $V_{CE} = 15\text{ V}$, and $I_C = 15\text{ mA}$.

- Determine whether this amplifier is unconditionally stable.
- Determine $G_{TU,max}$. Express your answer in dB.
- Determine the optimum source impedance and the optimum load impedance for maximum gain. Illustrate your answer in the Smith chart.
- Design a 2-element matching network at the input and a 2-element matching network at the output of the amplifier to reach conjugate matching to a $50\ \Omega$ source and load impedance. Illustrate your answer on the Smith chart.
- Calculate the constant gain circle for $G_L=1.5\text{ dB}$ and indicate it in the Smith chart.

Amplifier design

Similar as example 12.2 book of Pozar

The S parameters for the HP HFET-102 GaAs FET at 2 GHz with a bias voltage $V_{gs} = 0$ are given as follows ($Z_0 = 50 \Omega$):

$$S_{11} = 0.894 \angle -60.6^\circ,$$

$$S_{21} = 3.122 \angle 123.6^\circ,$$

$$S_{12} = 0.020 \angle 62.4^\circ,$$

$$S_{22} = 0.781 \angle -27.6^\circ.$$

Determine the stability of this transistor by calculating K and $|\Delta|$, and plot the stability circles.

Amplifier design

Example 11.2 book of Pozar - Solution

From (11.31) and (11.24) we compute K and $|\Delta|$ as

$$\Delta = S_{11}S_{22} - S_{12}S_{21} = 0.696 \angle -83^\circ,$$

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}S_{21}|} = 0.607.$$

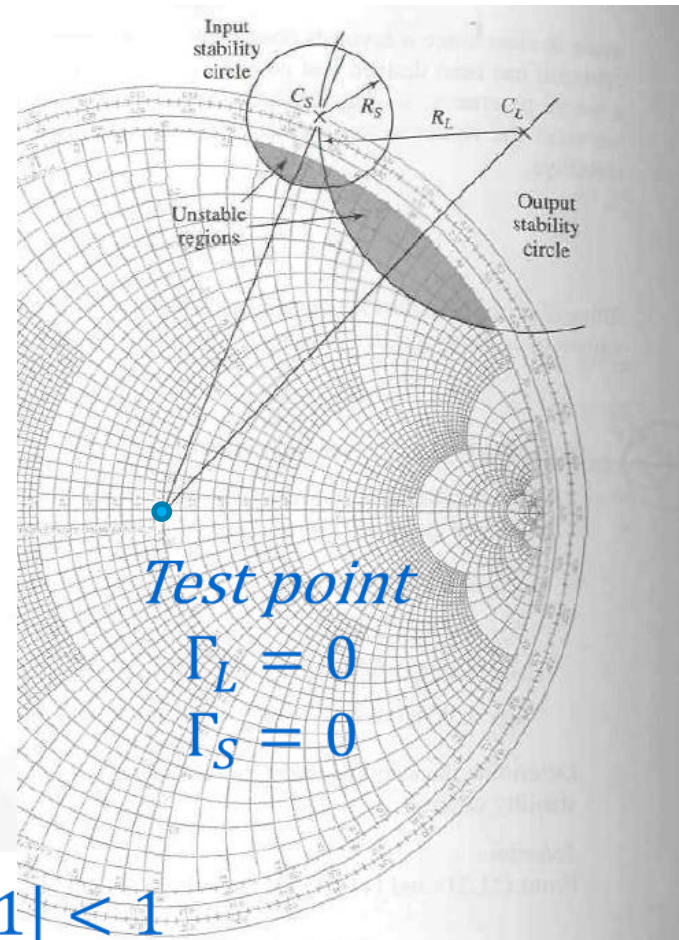
We have $|\Delta| = 0.696 < 1$, but $K < 1$, so the device is potentially unstable. The centers and radii of the stability circles are given by (11.28) and (11.29):

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2} = 1.361 \angle 47^\circ,$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2} = 0.50,$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2} = 1.132 \angle 68^\circ,$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2} = 0.199.$$



$$\Gamma_{in} = \frac{V_1^-}{V_1^+} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$$

$$\Gamma_{out} = \frac{V_2^-}{V_2^+} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S}$$

$$\Gamma_{in} = S_{11} \text{ for } \Gamma_L = 0, |S_{11}| < 1$$

$$\Gamma_{out} = S_{22} \text{ for } \Gamma_S = 0, |S_{22}| < 1$$

Amplifier design: problem 5

The S-parameters and the noise parameters of a transistor at 1 GHz are

$$S_{11} = 0.76 \angle -160^\circ$$

$$S_{12} = 0$$

$$S_{21} = 5.01 \angle 85^\circ$$

$$S_{22} = 0.508 \angle -20^\circ$$

$$F_{\min} = 2.2 \text{ dB}$$

$$\Gamma_{\text{opt}} = 0.5 \angle -150^\circ$$

$$r_n = 6.6 \Omega$$

- Proof that the transistor is unconditionally stable.
- How must the source reflection coefficient Γ_S and load reflection coefficient Γ_L be chosen to achieve maximum transducer power gain $G_{\text{TU,max}}$. Calculate the required Γ_S and Γ_L and indicate them on the Smith chart. What are the corresponding values of the source impedance Z_S and the load impedance Z_L ?
- For the case of conjugate matching at the input side of the transistor, design an input matching network consisting of an inductor and a capacitor. Calculate the value of the inductance L and the capacitance C required for this network. Illustrate your answer on the Smith chart.
- Determine the maximum transducer power gain $G_{\text{TU,max}}$. Give your answer in dB.
- For the case that the input matching network is not conjugately matched, calculate the constant gain circle that will lead to a 3 dB lower $G_{\text{TU,max}}$. Draw this constant gain circle on the Smith chart.
- Calculate the center and the radius of the 3 dB constant noise figure circle. Draw the 3 dB constant noise figure circle on the Smith chart.

Amplifier design: problem 5 – solution a)

a)

$$|\Delta| = |S_{11} S_{22} - S_{12} S_{21}| < 1$$
$$|\Delta| = |0.76 e^{-j160} \cdot 0.508 e^{-j20}|$$
$$|\Delta| = 0.386 < 1 \rightarrow \text{OK}$$

$K = \infty > 1 \rightarrow \text{OK}$

generic case

$$|S_{11}| < 1 \Rightarrow 0.76 < 1 \rightarrow \text{OK}$$
$$|S_{22}| < 1 \Rightarrow 0.508 < 1 \rightarrow \text{OK}$$

unilateral case

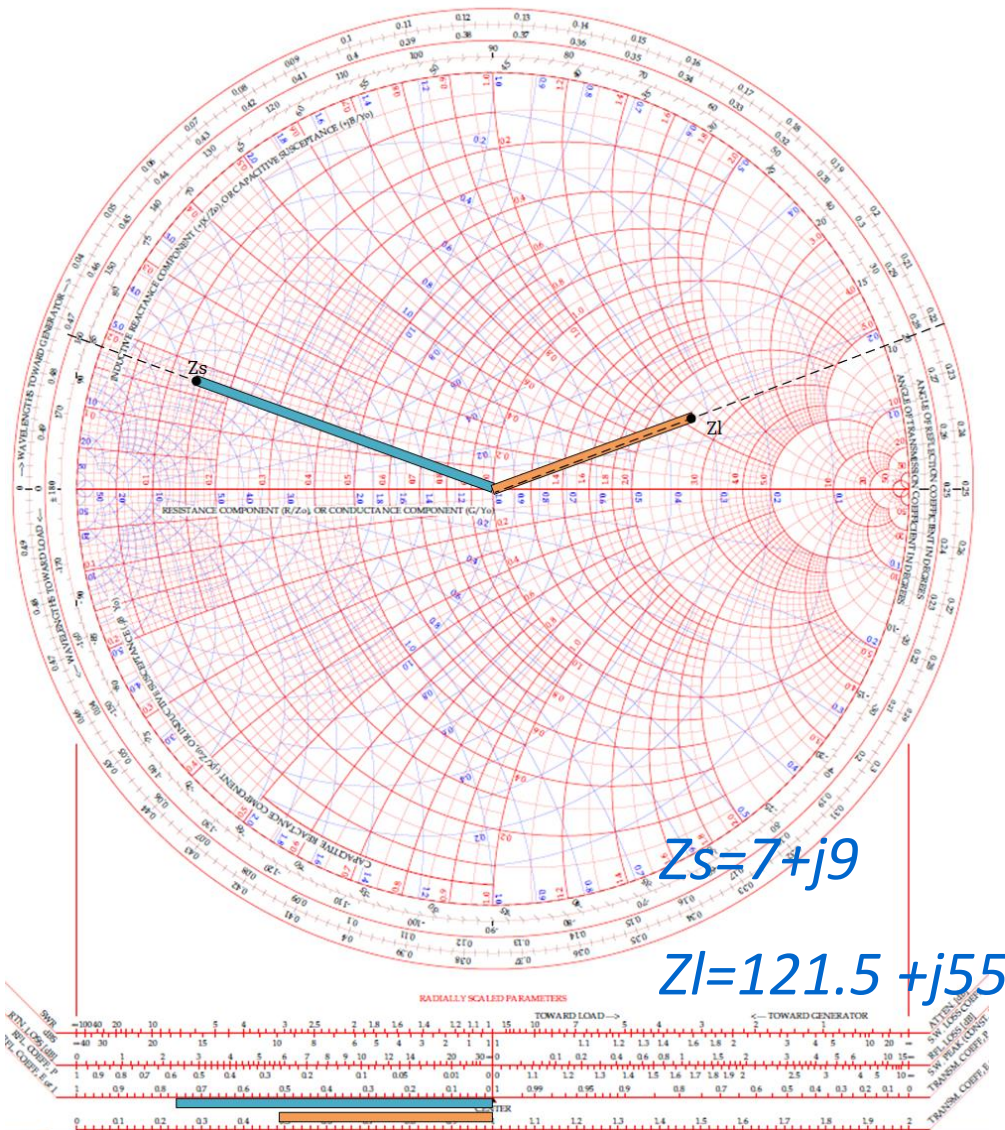
b) $\Gamma_{TU \max}$ / unilateral case $S_{12} = 0$

$$\Gamma_{TU \max} = \frac{1}{1 - |S_{11}|^2} |S_{21}|^2 \frac{1}{1 - |S_{22}|^2}$$

$$\Gamma_S = S_{11}^* = 0.76 e^{j160}$$

$$\Gamma_L = S_{22}^* = 0.508 e^{j20}$$

Amplifier design: problem 5 – solution b)



$$\Gamma_s = \frac{Z_s - Z_0}{Z_s + Z_0}$$

$$\Gamma_s (Z_s + Z_0) = Z_s - Z_0$$

$$\Gamma_s Z_s - Z_s = -Z_0 - \Gamma_s Z_0$$

$$Z_s (\Gamma_s - 1) = -Z_0 (1 + \Gamma_s)$$

$$Z_s = Z_0 \cdot \frac{1 + \Gamma_s}{1 - \Gamma_s}$$

$$Z_l = Z_0 \cdot \frac{1 + \Gamma_l}{1 - \Gamma_l}$$

$$Z_s = Z_0 \cdot \frac{1 + 0.16e^{j160}}{1 - 0.16e^{j160}} = Z_0 \cdot \frac{1 - 0.14 + j0.26}{1 + 0.14 - j0.26}$$

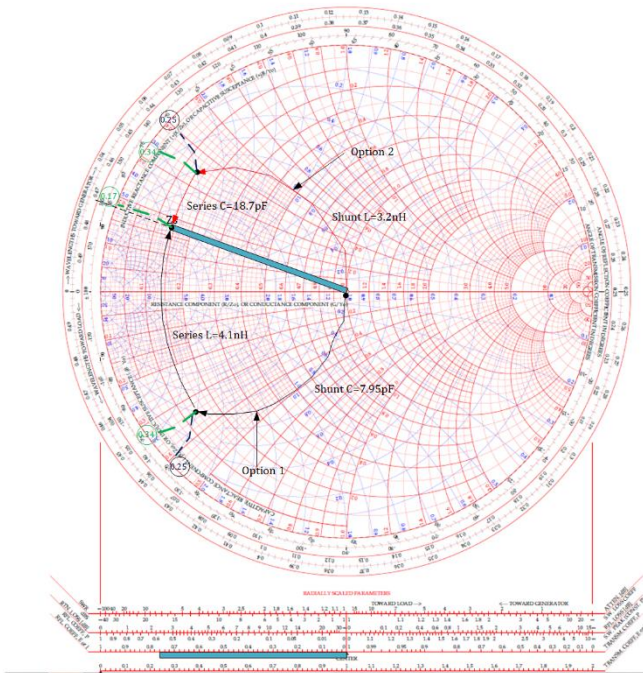
$$Z_s = Z_0 \cdot \frac{0.86 + j0.26}{1.71 - j0.26} = Z_0 \cdot \frac{1.71 + j0.26}{1.71 + j0.26}$$

$$Z_s = Z_0 \cdot \frac{0.42 + j0.52}{3} = Z_0 \cdot (0.14 + j0.18)$$

$$Z_l = Z_0 \cdot \frac{1 + 0.508e^{j20}}{1 - 0.508e^{j20}} = Z_0 \cdot \frac{1.47 + j0.7}{0.52 - j0.17}$$

$$Z_l = Z_0 \cdot \frac{0.73 + j0.33}{0.3} = Z_0 \cdot (2.43 + j1.1)$$

Amplifier design: problem 5 – solution c)



c) option 1

Diagram: A series inductor L followed by a shunt capacitor C connected to ground. The input impedance is Z_{in} and the source impedance is $Z_s = 50 \Omega$.

$$X_{in} = j \omega L = j \frac{2\pi f L}{0.02} \Rightarrow C = 7.95 \text{ pF}$$

$f = 1 \text{ GHz}$

$$Z_{in} = j (0.34 + j 1.8) = j \frac{2\pi f L}{50} \Rightarrow L = 4.1 \text{ nH}$$

c) option 2

Diagram: A shunt inductor L connected to ground, followed by a series capacitor C connected to the load Z_L . The input impedance is Z_{in} and the source impedance is $Z_s = 50 \Omega$.

$$Z_{in} = -j \frac{1}{\omega L} = -j \frac{1}{0.02 \omega L}$$

$$Z_{in} = -j 2.5 = -j \frac{1}{0.02 \cdot 2\pi \cdot 1 \text{ GHz} \cdot L}$$

$$L = 3.2 \text{ nH}$$

$$Z_{in} = -j \frac{1}{50 \omega C} = -j \frac{1}{50 \cdot 2\pi \cdot 1 \text{ GHz} \cdot C}$$

$$Z_{in} = +j (0.17 - 0.34) = -j \frac{1}{50 \cdot 2\pi \cdot 1 \text{ GHz} \cdot C}$$

$$C = 18.7 \text{ pF}$$

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Amplifier design: problem 5 – solution d)

$$G_{TU \max} = \frac{1}{1 - |S_{11}|^2} |S_{21}|^2 \frac{1}{1 - |S_{22}|^2}$$
$$G_{TU \max} = 2,36 \cdot 25,1 \cdot 1,39$$
$$G_{TU \max} = 79,4$$
$$[G_{TU \max}]_{dB} = 19 \text{ dB}$$

Amplifier design: problem 5 – solution e)

$$G_{s,max} = \frac{1}{1-|s_{11}|^2} = 2.36$$

$$G_{s,max} = 3.73 \text{ dB}$$

$$G_{s,required} = G_{s,max} - 3 \text{ dB}$$

$$G_{s,required} = 0.73 \text{ dB}$$

$$G_{s,required} = 1.18$$

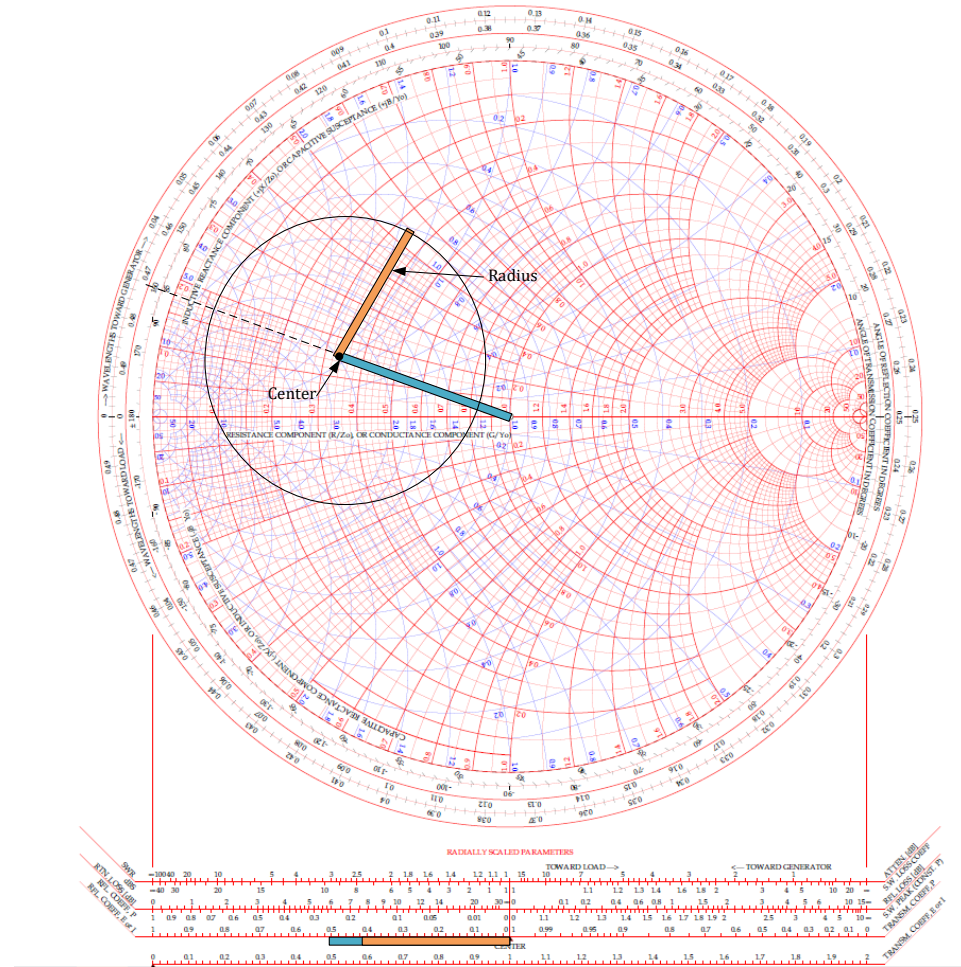
$$g_s = \frac{1.18}{2.36} = 0.5$$

$$C_s = \frac{0.5 \cdot |s_{11}|^2}{1 - (1 - g_s) |s_{11}|^2}$$

$$C_s = \frac{0.5 \cdot 0.76 \cdot e^{j160}}{1 - (1 - 0.5) \cdot 0.76^2} = 0.53 e^{j160}$$

$$R_s = \frac{\sqrt{1 - g_s} (1 - |s_{11}|^2)}{1 - (1 - g_s) |s_{11}|^2} = 0.42$$

$$R_s = \frac{0.71}{0.71}$$



Amplifier design: problem 5 – solution f)

$$f) \quad H = (F - F_{min}) \frac{1 + |\Gamma_{opt}|^2}{4r_n}$$

$$r_n = 6.6$$

$$\Gamma_{opt} = 0.5 e^{-j150}$$

$$F_{min} = 1.2 \text{ dB} \Rightarrow F_{min} = 1.66$$

$$F = 3 \text{ dB} \Rightarrow F = 2$$

$$H = (2 - 1.66) \frac{1 + 0.5 \cos 150 - j0.584}{4 \cdot 6.6}$$

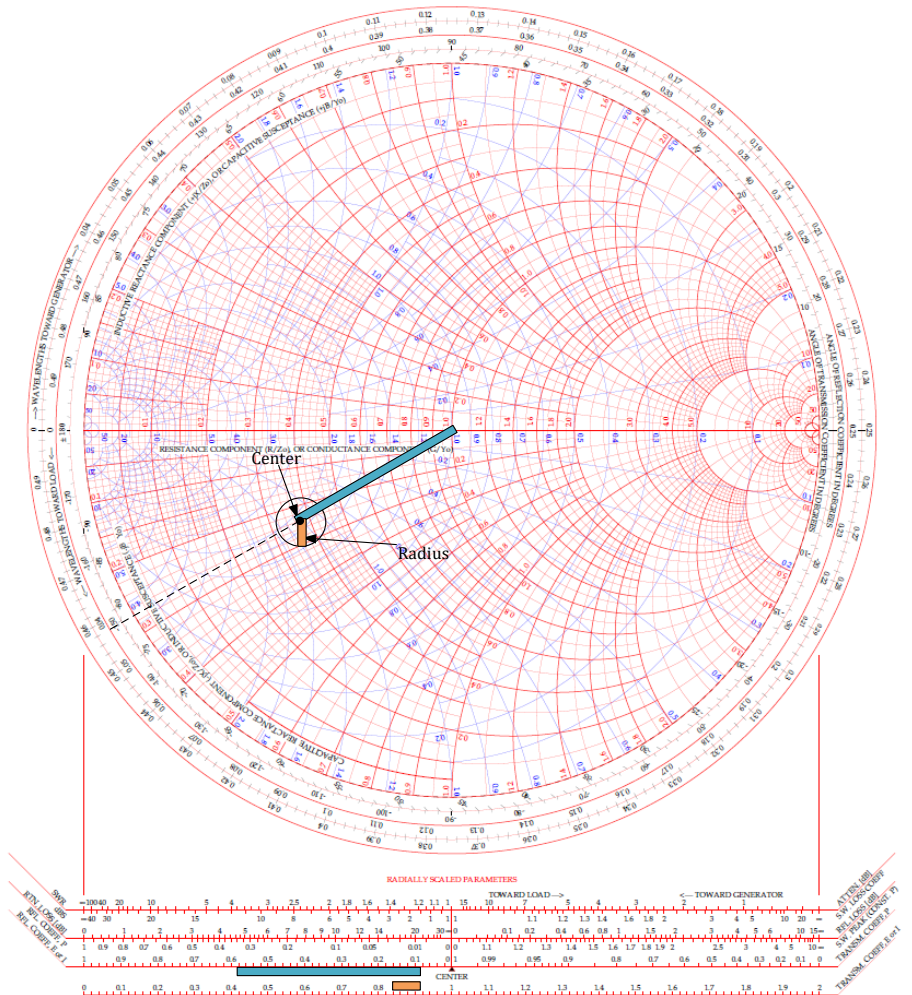
$$H = \frac{0.34 \cdot |0.56 - j0.25|^2}{4 \cdot 6.6}$$

$$H = \frac{0.34 \cdot 0.38}{4 \cdot 6.6} = 4.8 \cdot 10^{-3}$$

$$C_F = \frac{\Gamma_{opt}}{1 + H} = \frac{0.497 e^{-j150}}{1 + H}$$

$$R_F = \frac{1}{H + 1} \sqrt{H^2 + H(1 - |\Gamma_{opt}|^2)}$$

$$\underline{R_F = 0.06}$$



Example 11.5 book of Pozar

A GaAs FET is biased for minimum noise figure, and has the following S parameters and noise parameters at 4 GHz ($Z_0 = 50\ \Omega$): $S_{11} = 0.6\angle -60^\circ$, $S_{21} = 1.9\angle 81^\circ$, $S_{12} = 0.05\angle 26^\circ$, $S_{22} = 0.5\angle -60^\circ$; $F_{\min} = 1.6\text{ dB}$, $\Gamma_{\text{opt}} = 0.62\angle 100^\circ$, $R_N = 20\ \Omega$. For design purposes, assume the device is unilateral, and calculate the maximum error in G_T resulting from this assumption. Then design an amplifier having a 2.0 dB noise figure with the maximum gain that is compatible with this noise figure.

Example 11.4 book of Pozar – solution (1/2)

Next, we use (11.59) and (11.61) to compute the center and radius of the 2 dB noise figure circle:

$$N = \frac{F - F_{\min}}{4R_N/Z_0} |1 + \Gamma_{\text{opt}}|^2 = \frac{1.58 - 1.445}{4(20/50)} |1 + 0.62 \angle 100^\circ|^2$$

$$= 0.0986,$$

$$C_F = \frac{\Gamma_{\text{opt}}}{N + 1} = 0.56 \angle 100^\circ$$

$$R_F = \frac{\sqrt{N(N + 1 - |\Gamma_{\text{opt}}|^2)}}{N + 1} = 0.24.$$

This noise figure circle is plotted in Figure 11.15a. Minimum noise figure ($F_{\min} = 1.6$ dB) occurs for $\Gamma_S = \Gamma_{\text{opt}} = 0.62 \angle 100^\circ$.

Next we calculate data for several input section constant gain circles. From (11.52),

G_S (dB)	g_S	C_S	R_S
1.0	0.805	$0.52 \angle 60^\circ$	0.300
1.5	0.904	$0.56 \angle 60^\circ$	0.205
1.7	0.946	$0.58 \angle 60^\circ$	0.150

These circles are also plotted in Figure 11.15a. We see that the $G_S = 1.7$ dB gain circle just intersects the $F = 2$ dB noise figure circle, and that any higher gain will result in a worse noise figure. From the Smith chart the optimum solution is then $\Gamma_S = 0.53 \angle 75^\circ$, yielding $G_S = 1.7$ dB and $F = 2.0$ dB.

For the output section we choose $\Gamma_L = S_{22}^* = 0.5 \angle 60^\circ$ for a maximum G_L of

$$G_L = \frac{1}{1 - |S_{22}|^2} = 1.33 \approx 1.25 \text{ dB}.$$

The transistor gain is

$$G_0 = |S_{21}|^2 = 3.61 = 5.58 \text{ dB},$$

